Cambridge (CIE) IGCSE Computer Science

Your notes

Boolean Logic

Contents

Logic Gates

Your notes

Logic gates

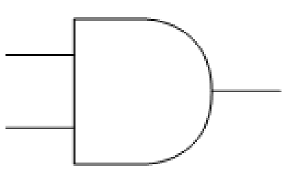
What is Boolean logic?

- Boolean logic is used in computer science and electronics to make **logical decisions**
- Boolean operators are either **TRUE** or **FALSE**, often represented as **1** or **0**
- Inputs and outputs are given **letters to represent them**
- To define Boolean logic we use **special symbols** to make **writing expressions** much **easier**

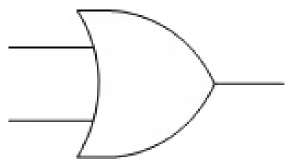
What are logic gates?

- Logic gates are a **visual way** of representing a **Boolean expression**
- The logic gates covered in this course are:
  - **AND**
  - **OR**
  - **NOT**
  - **XOR**
  - **NAND**
  - **NOR**

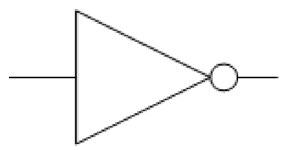
AND

Expression operator	Circuit symbol	Explanation
(A **AND** B)		Returns TRUE only if **both** inputs are TRUE **TRUE** AND **TRUE** = **TRUE** Otherwise = FALSE

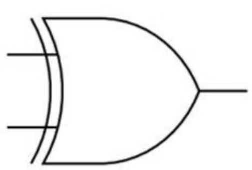
OR

Expression operator	Circuit symbol	Explanation
(A **OR** B)		Returns TRUE if **either** input is TRUE **TRUE** OR **FALSE** = **TRUE** **FALSE** OR **FALSE** = **FALSE**

NOT

Expression operator	Circuit symbol	Explanation
(**NOT** A)		**Reverses** the input value NOT **TRUE** = **FALSE** NOT **FALSE** = **TRUE**

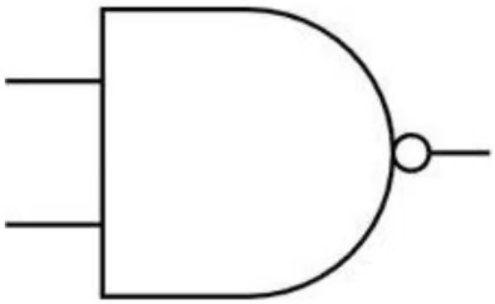
XOR (exclusive OR)

Expression operator	Circuit symbol	Explanation
(A **XOR** B)		Returns TRUE if **either** input is TRUE but **NOT** both **TRUE** OR **FALSE** = **TRUE** **FALSE** OR **FALSE** = **FALSE** **TRUE** OR **TRUE** = **FALSE**

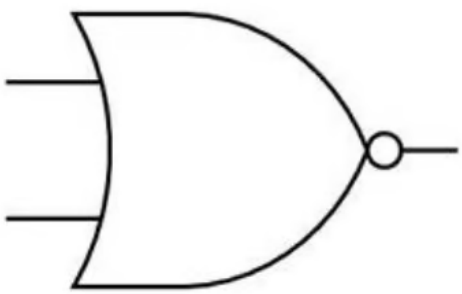
NAND (not and)

Expression operator	Circuit symbol	Explanation

Your notes

(A **NAND** B)		Returns TRUE if **either or both** inputs are FALSE **TRUE** OR **FALSE** = **TRUE** **FALSE** OR **FALSE** = **TRUE** **TRUE** OR **TRUE** = **FALSE**

NOR (not OR)

Expression operator	**Circuit symbol**	**Explanation**
(A **NOR** B)		Returns TRUE if **both** inputs are FALSE **TRUE** OR **FALSE** = **FALSE** **FALSE** OR **FALSE** = **TRUE** **TRUE** OR **TRUE** = **FALSE**

Examiner Tips and Tricks

Your notes

You will need to either draw a diagram of a logic circuit using these symbols, or you will have to write the boolean expression from an existing diagram.

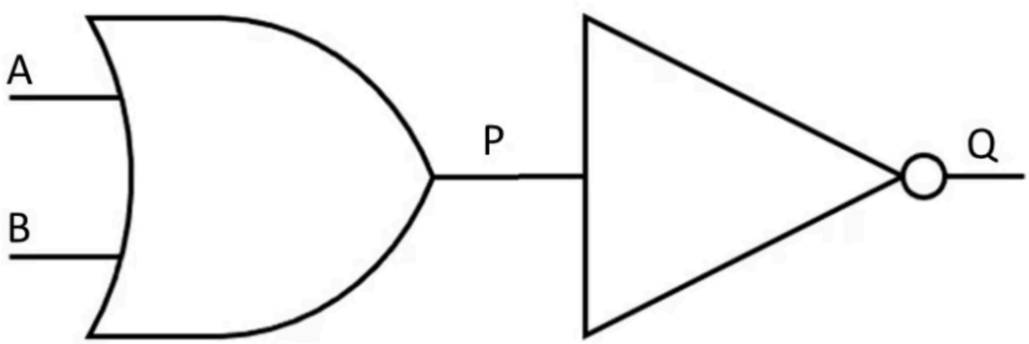
Logic Circuits

Your notes

Logic Circuits

What is a logic circuit?

- A logic circuit **performs logical operations** on binary information
- Logic gates are **core components** of logic circuits
- Based on the **arrangement of logic gates** and the **input signals**, the circuit performs a specific function
- A logic diagram is a **visual representation** of a combinations of logic gates within a logic circuit
- **Brackets** are used to clarify the **order** of operations
- An example logic diagram for the Boolean expression **Q = NOT(A OR B)** would be:



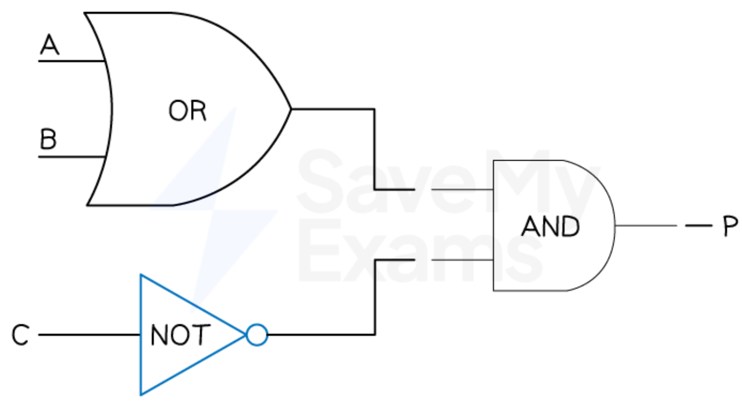
- In the logic diagram above, the **inputs** are represented by **A and B**
- **P** is the **output** of the OR gate on the left and becomes the input of the NOT gate
- **Q** is the **final output** of the logic circuit

Examiner Tips and Tricks

You may be asked to draw a logic circuit from a logic statement or a Boolean expression **OR** write the logical expression that is expressed in the logic diagram using Boolean expression operators

Logic circuits will be limited to a **maximum of three inputs and one output**

Example of combining logic gates



- **P = (A OR B) AND NOT C**

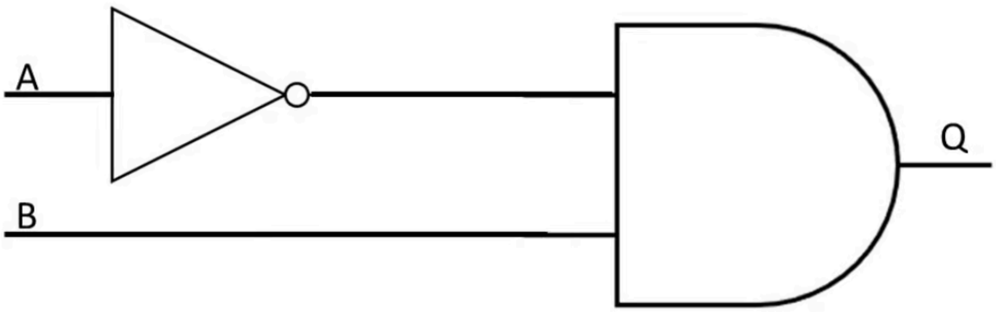
Worked Example

A sprinkler system switches on if it is not daytime (input A) and the temperature is greater than 40 (input B)

Draw a logic circuit to represent the problem statement above

[2]

Your notes


A
B
Q

Your notes

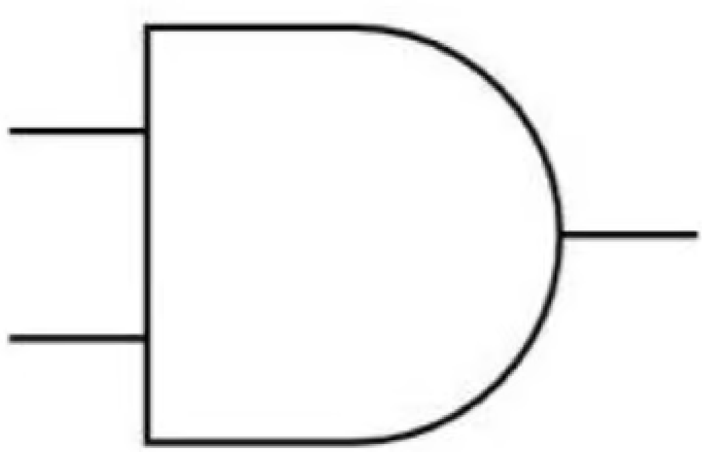
Truth Tables

Truth Tables

What is a truth table?

- A truth table is a tool used in logic and computer science to **visualise** the **results** of **Boolean expressions**
- They represent **all possible inputs** and the **associated outputs** for a **given Boolean expression**

AND

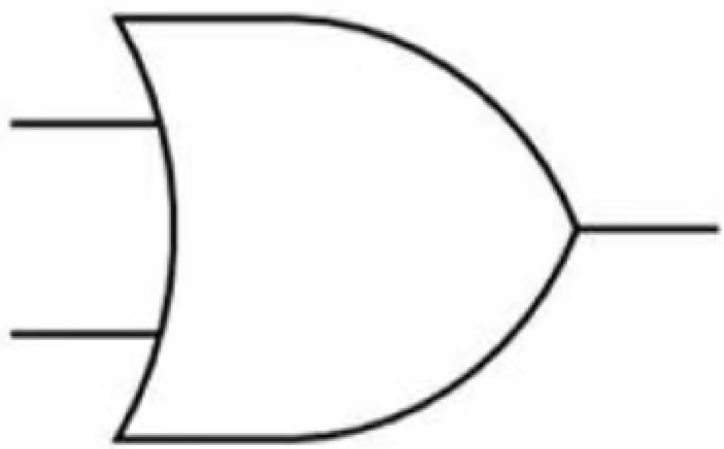
<table>
<tr><th>Circuit symbol</th><th>Truth Table</th></tr>
<tr><td></td><td>

A	B	A AND B
0	0	0
0	1	0
1	0	0
1	1	1

</td></tr>
</table>

OR

Circuit symbol	Truth Table

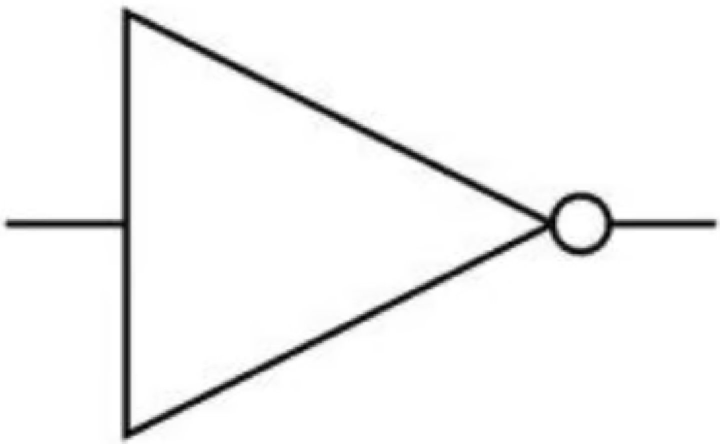


A	B	A OR B
0	0	0
0	1	1
1	0	1
1	1	1

Your notes

NOT

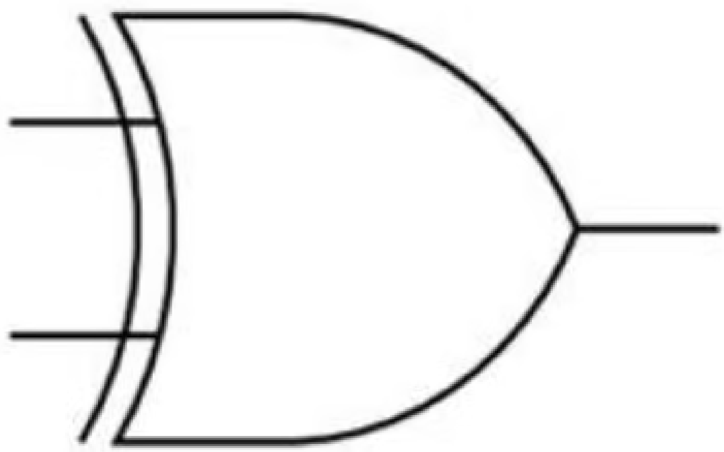
Circuit symbol	Truth Table



A	NOT
0	1
1	0

XOR (exclusive)

Circuit symbol	Truth Table



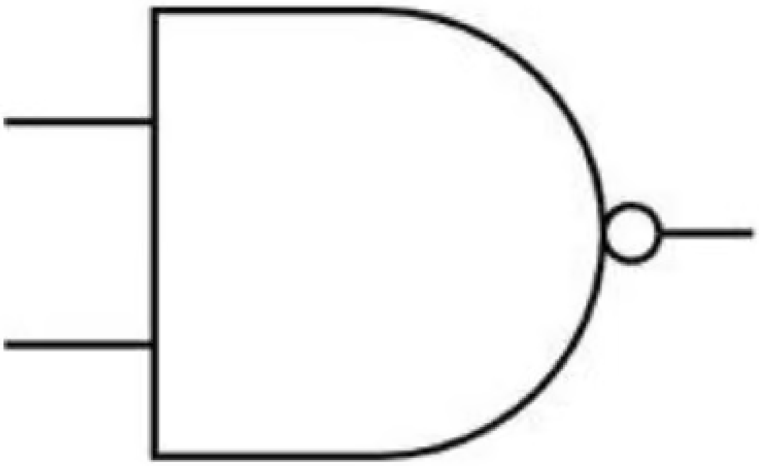
A	B	A XOR B
0	0	0
0	1	1
1	0	1
1	1	0

Your notes

NAND (not and)

Circuit symbol	Truth Table

Your notes

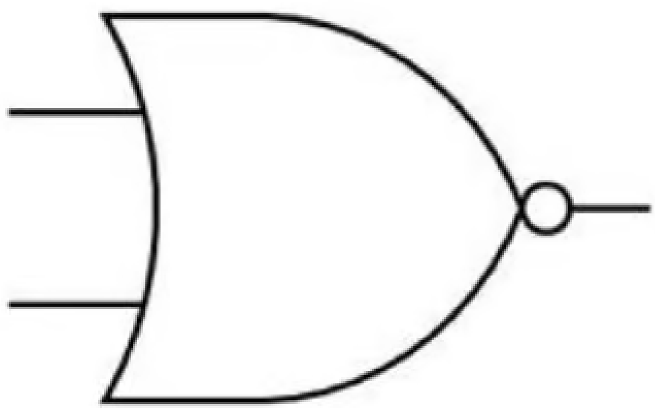
	A | B | NOT (A AND B) 0 | 0 | 1 0 | 1 | 1 1 | 0 | 1 1 | 1 | 0

A	B	NOT (A AND B)
0	0	1
0	1	1
1	0	1
1	1	0

NOR (not or)

Circuit symbol	Truth Table

Your notes



A	B	NOT (A OR B)
0	0	1
0	1	0
1	0	0
1	1	0

Truth Tables for Logic Circuits

How do you create truth tables for logic circuits?

- To create a truth table for the expression **P = (A AND B) AND NOT C**
  - Calculate the numbers of rows needed (**$2^{\text{number of inputs}}$**)
  - In this example there are 3 inputs (**A, B, C**) so a total of **8 rows** are needed (**2^3**)
  - To not miss any combination of inputs, start with **000** and **count** up in **3-bit binary (0-7)**

A	B	C
0	0	0
0	0	1
0	1	0
0	1	1

1	0	0
1	0	1
1	1	0
1	1	1

- Add a new column to show the **results** of the brackets first (A AND B)

A	B	C	A AND B
0	0	0	**0**
0	0	1	**0**
0	1	0	**0**
0	1	1	**0**
1	0	0	**0**
1	0	1	**0**
1	1	0	**1**
1	1	1	**1**

- Add a new column to show the **results** of NOT C

A	B	C	A AND B	NOT C
0	0	0	0	**1**
0	0	1	0	**0**

Your notes

0	1	0	0	**1**
0	1	1	0	**0**
1	0	0	0	**1**
1	0	1	0	**0**
1	1	0	1	**1**
1	1	1	1	**0**

Your notes

- The last column shows the **result** of the Boolean expression (**P**) by comparing (A AND B) AND NOT C // $(A.B).\overline{C}$

A	B	C	A AND B	NOT C	P
0	0	0	0	1	**0**
0	0	1	0	0	**0**
0	1	0	0	1	**0**
0	1	1	0	0	**0**
1	0	0	0	1	**0**
1	0	1	0	0	**0**
1	1	0	1	1	**1**
1	1	1	1	0	**0**

Examiner Tips and Tricks

Your notes

It is possible to create a truth table when combining expressions that show only the inputs and the final outputs.

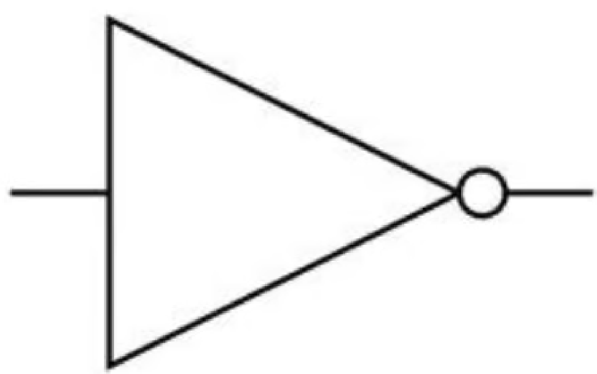
The inclusion of the extra columns supports the process but can be skipped if you feel able to do those in your head as you go.

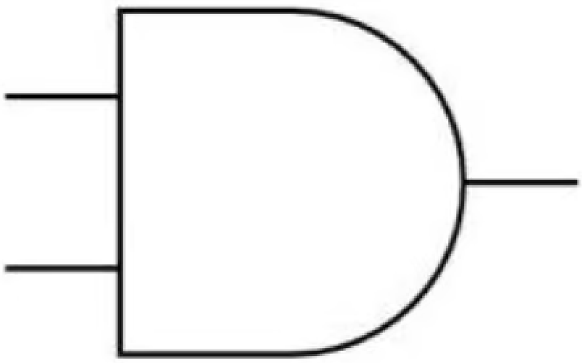
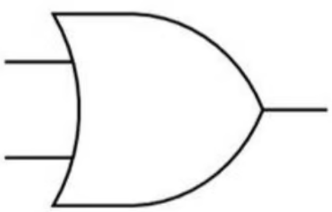
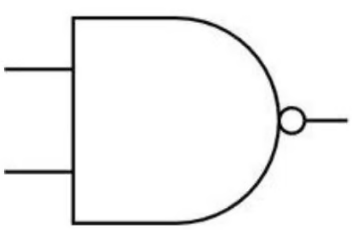
Logic Expressions

Your notes

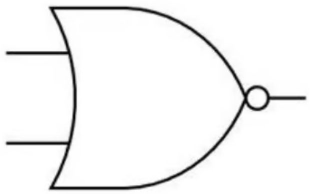
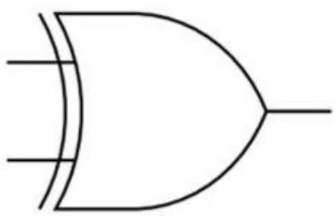
Logic Expressions

- A **logic expression** is a way of expressing a logic gate or logic circuit as an **equation**
- The **output** appears on the **left** of the equals sign with the **inputs** and **logic gates** on the **right**

<table>
<tr><th>Gate</th><th>Symbol</th><th>Truth Table</th><th>Logic Expression</th></tr>
<tr><td>NOT</td><td></td><td><table><tr><th>A</th><th>Z</th></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td></tr></table></td><td>Z = NOT A</td></tr>
</table>

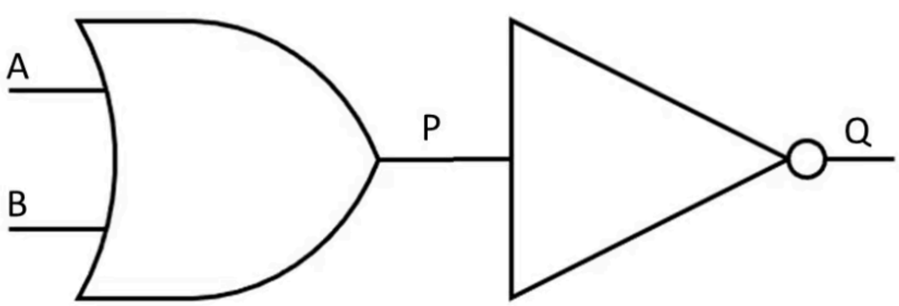
<table>
<tr><td>AND</td><td></td><td>
<table>
<tr><th>A</th><th>B</th><th>Z</th></tr>
<tr><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>1</td><td>1</td></tr>
</table>
</td><td>Z = A AND B</td></tr>
<tr><td>OR</td><td></td><td>
<table>
<tr><th>A</th><th>B</th><th>Z</th></tr>
<tr><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>0</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>1</td></tr>
</table>
</td><td>Z = A OR B</td></tr>
<tr><td>NAND</td><td></td><td>
<table>
<tr><th>A</th><th>B</th><th>Z</th></tr>
<tr><td>0</td><td>0</td><td>1</td></tr>
<tr><td>0</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>0</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>0</td></tr>
</table>
</td><td>Z = A NAND B</td></tr>
</table>

Your notes

NOR		**A** **B** **Z** 0 0 1 0 1 0 1 0 0 1 1 0	Z = A NOR B
XOR		**A** **B** **Z** 0 0 0 0 1 1 1 0 1 1 1 0	Z = A XOR B

- Logic circuits containing multiple gates can also be expressed as logic expressions/statements

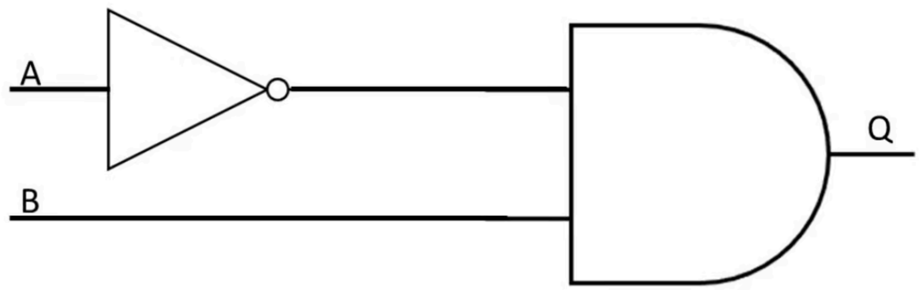
An example logic circuit containing two inputs



- The logic circuit above can be expressed as the logic expression **Q= NOT(A OR B)**

Your notes

An example logic circuit containing two inputs

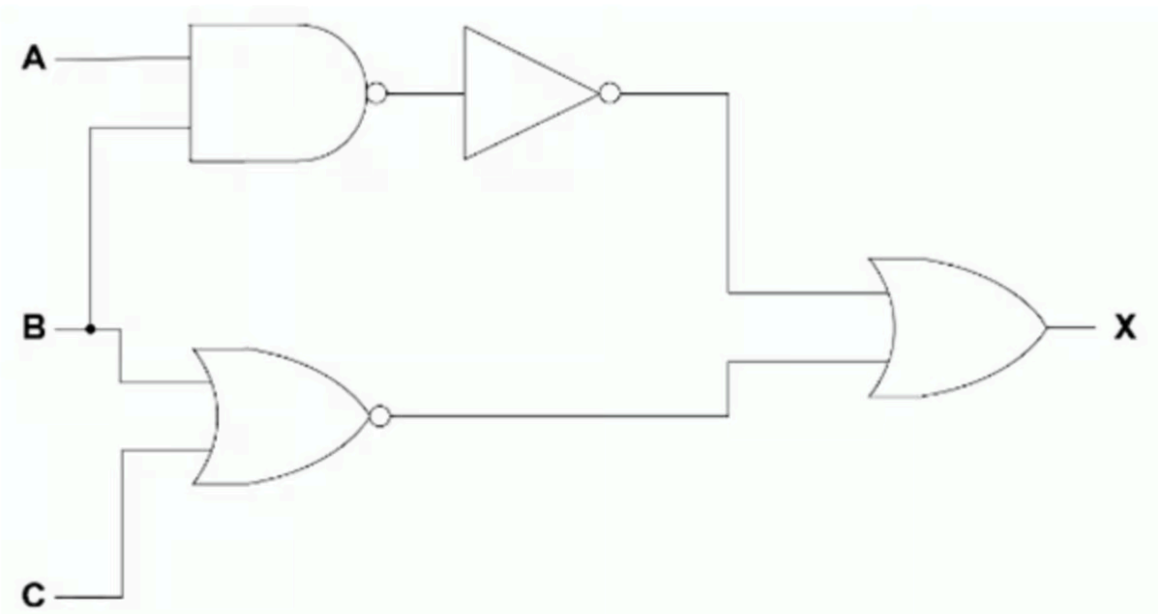


- The logic circuit above can be expressed as the logic expression **Q= (NOT A) AND B**

An example logic circuit containing three inputs

- The logic circuit above can be expressed as the logic expression **P = ((NOT A) OR B) NAND C**

An example logic circuit containing three inputs



- This logic circuit above can be expressed as **X = NOT (A NAND B) OR (B NOR C)**

Your notes

Examiner Tips and Tricks

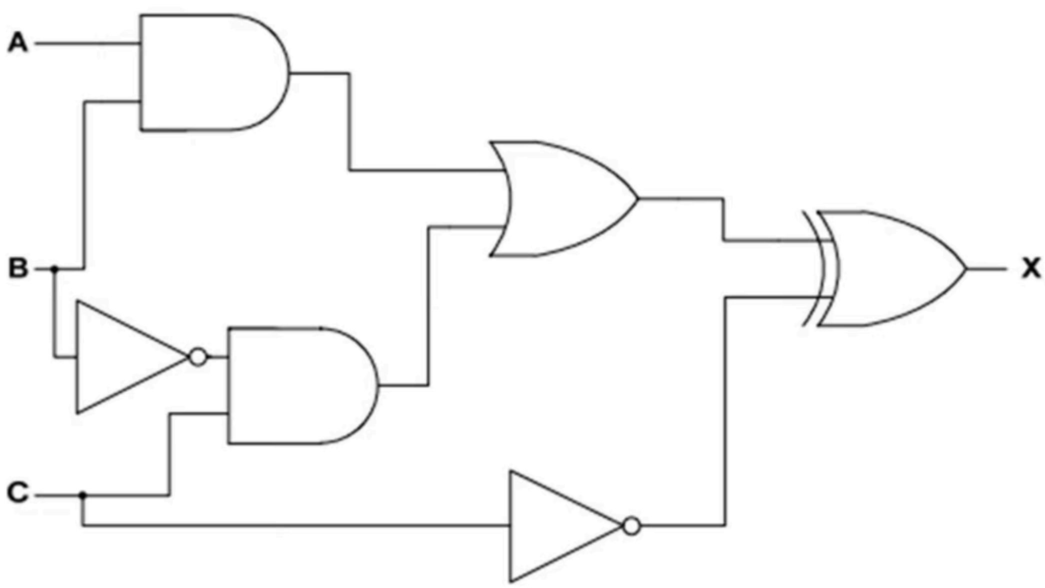
- You may be required to write a logic expression/statement from a **truth table** or a **logic circuit**. You may also have to do the opposite - **draw a logic circuit** and **complete a truth table for a logic expression**

Worked Example

Consider the logic statement: **X = (((A AND B) OR (C AND NOT B)) XOR NOT C)**

a. Draw a logic circuit to represent the given logic statement. **[6]**

Answer



One mark per correct logic gate, with the correct input